MSG No. 202.22 $Fm\bar{3}$ [Type I, cubic]

Table 1: Wyckoff site: `4a`, site symmetry: `m-3.`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]

Table 2: Wyckoff site: `4b`, site symmetry: `m-3.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2}, 0, 0\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]
3	$\left[0, \frac{1}{2}, 0\right]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
4	$\left[0, 0, \frac{1}{2}\right]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]

Table 3: Wyckoff site: `8c`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,6,26,34,35,51,55,60,76,80,81]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,10,11,25,29,30,52,56,57,75,79,84]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,7,12,28,32,33,49,53,54,74,82,83]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,8,9,27,31,36,50,58,59,73,77,78]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[13,17,18,38,46,47,63,67,72,88,92,93]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[14,22,23,37,41,42,64,68,69,87,91,96]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[15,19,24,40,44,45,61,65,66,86,94,95]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[16,20,21,39,43,48,62,70,71,85,89,90]

Table 4: Wyckoff site: `24d`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[1,14,26,37]
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[2,13,25,38]
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[3,16,28,39]
4	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[4,15,27,40]
5	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[5,24,60,65]
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[6,21,81,90]
7	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[7,23,83,91]
8	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[8,22,58,68]
9	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[9,18,78,93]
10	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[10,20,56,70]
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[11,19,79,95]
12	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[12,17,53,72]
13	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[29,48,84,89]

continued ...

Table 4

No.	position	mapping
14	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[30,45,57,66]
15	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[31,47,59,67]
16	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[32,46,82,92]
17	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[33,42,54,69]
18	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[34,44,80,94]
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[35,43,55,71]
20	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[36,41,77,96]
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,62,74,85]
22	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[50,61,73,86]
23	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[51,64,76,87]
24	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[52,63,75,88]

Table 5: Wyckoff site: 24e, site symmetry: 2mm..

No.	position	mapping
1	$[x, 0, 0]$	[1,2,15,16]
2	$[-x, 0, 0]$	[3,4,13,14]
3	$[0, x, 0]$	[5,12,20,22]
4	$[0, 0, x]$	[6,9,19,23]
5	$[0, 0, -x]$	[7,11,18,21]
6	$[0, -x, 0]$	[8,10,17,24]
7	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,39,40]
8	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[27,28,37,38]
9	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[29,36,44,46]
10	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[30,33,43,47]
11	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[31,35,42,45]
12	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[32,34,41,48]
13	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[49,50,63,64]
14	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[51,52,61,62]
15	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[53,60,68,70]
16	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[54,57,67,71]
17	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[55,59,66,69]
18	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[56,58,65,72]
19	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[73,74,87,88]
20	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[75,76,85,86]
21	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[77,84,92,94]
22	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[78,81,91,95]
23	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[79,83,90,93]
24	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[80,82,89,96]

Table 6: Wyckoff site: 32f, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,5,6]

continued ...

Table 6

No.	position	mapping
2	$[x,\ -x,\ -x]$	[2,10,11]
3	$[-x,\ x,\ -x]$	[3,7,12]
4	$[-x,\ -x,\ x]$	[4,8,9]
5	$[-x,\ -x,\ -x]$	[13,17,18]
6	$[-x,\ x,\ x]$	[14,22,23]
7	$[x,\ -x,\ x]$	[15,19,24]
8	$[x,\ x,\ -x]$	[16,20,21]
9	$\left[x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[25,29,30]
10	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[26,34,35]
11	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[27,31,36]
12	$\left[-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[28,32,33]
13	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[37,41,42]
14	$\left[-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[38,46,47]
15	$\left[x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[39,43,48]
16	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[40,44,45]
17	$\left[x+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$	[49,53,54]
18	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-x\right]$	[50,58,59]
19	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	[51,55,60]
20	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	[52,56,57]
21	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-x\right]$	[61,65,66]
22	$\left[\frac{1}{2}-x,\ x,\ x+\frac{1}{2}\right]$	[62,70,71]
23	$\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	[63,67,72]
24	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-x\right]$	[64,68,69]
25	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[73,77,78]
26	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	[74,82,83]
27	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -x\right]$	[75,79,84]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	[76,80,81]
29	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -x\right]$	[85,89,90]
30	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x\right]$	[86,94,95]
31	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x\right]$	[87,91,96]
32	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	[88,92,93]

Table 7: Wyckoff site: 48g, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,26]
2	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[2,25]
3	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,28]
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4,27]
5	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	[5,60]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	[6,81]
7	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[7,83]
8	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[8,58]
9	$\left[\frac{3}{4},\ \frac{3}{4},\ x\right]$	[9,78]
10	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	[10,56]

continued ...

Table 7

No.	position	mapping
11	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[11,79]
12	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	[12,53]
13	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[13,38]
14	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	[14,37]
15	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[15,40]
16	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[16,39]
17	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[17,72]
18	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[18,93]
19	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[19,95]
20	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[20,70]
21	$\left[\frac{1}{4}, \frac{1}{4}, -x\right]$	[21,90]
22	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[22,68]
23	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[23,91]
24	$\left[\frac{1}{4}, -x, \frac{1}{4}\right]$	[24,65]
25	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[29,84]
26	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[30,57]
27	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[31,59]
28	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[32,82]
29	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[33,54]
30	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[34,80]
31	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[35,55]
32	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[36,77]
33	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[41,96]
34	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[42,69]
35	$\left[\frac{1}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[43,71]
36	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[44,94]
37	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[45,66]
38	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[46,92]
39	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[47,67]
40	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[48,89]
41	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,74]
42	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[50,73]
43	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[51,76]
44	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	[52,75]
45	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[61,86]
46	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[62,85]
47	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[63,88]
48	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[64,87]

Table 8: Wyckoff site: 48h, site symmetry: m..

No.	position	mapping
1	$[0, y, z]$	[1,14]
2	$[0, -y, -z]$	[2,13]
3	$[0, y, -z]$	[3,16]

continued ...

Table 8

No.	position	mapping
4	$[0, -y, z]$	[4,15]
5	$[z, 0, y]$	[5,24]
6	$[y, z, 0]$	[6,21]
7	$[-y, z, 0]$	[7,23]
8	$[-z, 0, y]$	[8,22]
9	$[-y, -z, 0]$	[9,18]
10	$[z, 0, -y]$	[10,20]
11	$[y, -z, 0]$	[11,19]
12	$[-z, 0, -y]$	[12,17]
13	$\left[0, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[25,38]
14	$\left[0, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[26,37]
15	$\left[0, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[27,40]
16	$\left[0, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[28,39]
17	$\left[z, \frac{1}{2}, y+\frac{1}{2}\right]$	[29,48]
18	$\left[y, z+\frac{1}{2}, \frac{1}{2}\right]$	[30,45]
19	$\left[-y, z+\frac{1}{2}, \frac{1}{2}\right]$	[31,47]
20	$\left[-z, \frac{1}{2}, y+\frac{1}{2}\right]$	[32,46]
21	$\left[-y, \frac{1}{2}-z, \frac{1}{2}\right]$	[33,42]
22	$\left[z, \frac{1}{2}, \frac{1}{2}-y\right]$	[34,44]
23	$\left[y, \frac{1}{2}-z, \frac{1}{2}\right]$	[35,43]
24	$\left[-z, \frac{1}{2}, \frac{1}{2}-y\right]$	[36,41]
25	$\left[\frac{1}{2}, y, z+\frac{1}{2}\right]$	[49,62]
26	$\left[\frac{1}{2}, -y, \frac{1}{2}-z\right]$	[50,61]
27	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	[51,64]
28	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[52,63]
29	$\left[z+\frac{1}{2}, 0, y+\frac{1}{2}\right]$	[53,72]
30	$\left[y+\frac{1}{2}, z, \frac{1}{2}\right]$	[54,69]
31	$\left[\frac{1}{2}-y, z, \frac{1}{2}\right]$	[55,71]
32	$\left[\frac{1}{2}-z, 0, y+\frac{1}{2}\right]$	[56,70]
33	$\left[\frac{1}{2}-y, -z, \frac{1}{2}\right]$	[57,66]
34	$\left[z+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	[58,68]
35	$\left[y+\frac{1}{2}, -z, \frac{1}{2}\right]$	[59,67]
36	$\left[\frac{1}{2}-z, 0, \frac{1}{2}-y\right]$	[60,65]
37	$\left[\frac{1}{2}, y+\frac{1}{2}, z\right]$	[73,86]
38	$\left[\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[74,85]
39	$\left[\frac{1}{2}, y+\frac{1}{2}, -z\right]$	[75,88]
40	$\left[\frac{1}{2}, \frac{1}{2}-y, z\right]$	[76,87]
41	$\left[z+\frac{1}{2}, \frac{1}{2}, y\right]$	[77,96]
42	$\left[y+\frac{1}{2}, z+\frac{1}{2}, 0\right]$	[78,93]
43	$\left[\frac{1}{2}-y, z+\frac{1}{2}, 0\right]$	[79,95]
44	$\left[\frac{1}{2}-z, \frac{1}{2}, y\right]$	[80,94]
45	$\left[\frac{1}{2}-y, \frac{1}{2}-z, 0\right]$	[81,90]
46	$\left[z+\frac{1}{2}, \frac{1}{2}, -y\right]$	[82,92]
47	$\left[y+\frac{1}{2}, \frac{1}{2}-z, 0\right]$	[83,91]
48	$\left[\frac{1}{2}-z, \frac{1}{2}, -y\right]$	[84,89]

Table 9: Wyckoff site: `96i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x,\ -y,\ -z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[y,\ z,\ x]$	`[6]`
7	$[-y,\ z,\ -x]$	`[7]`
8	$[-z,\ -x,\ y]$	`[8]`
9	$[-y,\ -z,\ x]$	`[9]`
10	$[z,\ -x,\ -y]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-z,\ x,\ -y]$	`[12]`
13	$[-x,\ -y,\ -z]$	`[13]`
14	$[-x,\ y,\ z]$	`[14]`
15	$[x,\ -y,\ z]$	`[15]`
16	$[x,\ y,\ -z]$	`[16]`
17	$[-z,\ -x,\ -y]$	`[17]`
18	$[-y,\ -z,\ -x]$	`[18]`
19	$[y,\ -z,\ x]$	`[19]`
20	$[z,\ x,\ -y]$	`[20]`
21	$[y,\ z,\ -x]$	`[21]`
22	$[-z,\ x,\ y]$	`[22]`
23	$[-y,\ z,\ x]$	`[23]`
24	$[z,\ -x,\ y]$	`[24]`
25	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[25]`
26	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[26]`
27	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[27]`
28	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[28]`
29	$\left[z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[29]`
30	$\left[y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[30]`
31	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[31]`
32	$\left[-z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	`[32]`
33	$\left[-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	`[33]`
34	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	`[34]`
35	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	`[35]`
36	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[36]`
37	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[37]`
38	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[38]`
39	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[39]`
40	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[40]`
41	$\left[-z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	`[41]`
42	$\left[-y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	`[42]`
43	$\left[y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	`[43]`
44	$\left[z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[44]`
45	$\left[y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[45]`
46	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[46]`

continued ...

Table 9

No.	position	mapping
47	$\left[-y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[47]
48	$\left[z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[48]
49	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[49]
50	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[50]
51	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[51]
52	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[52]
53	$\left[z+\frac{1}{2},\, x,\, y+\frac{1}{2}\right]$	[53]
54	$\left[y+\frac{1}{2},\, z,\, x+\frac{1}{2}\right]$	[54]
55	$\left[\frac{1}{2}-y,\, z,\, \frac{1}{2}-x\right]$	[55]
56	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	[56]
57	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	[57]
58	$\left[z+\frac{1}{2},\, -x,\, \frac{1}{2}-y\right]$	[58]
59	$\left[y+\frac{1}{2},\, -z,\, \frac{1}{2}-x\right]$	[59]
60	$\left[\frac{1}{2}-z,\, x,\, \frac{1}{2}-y\right]$	[60]
61	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[61]
62	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[62]
63	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[63]
64	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[64]
65	$\left[\frac{1}{2}-z,\, -x,\, \frac{1}{2}-y\right]$	[65]
66	$\left[\frac{1}{2}-y,\, -z,\, \frac{1}{2}-x\right]$	[66]
67	$\left[y+\frac{1}{2},\, -z,\, x+\frac{1}{2}\right]$	[67]
68	$\left[z+\frac{1}{2},\, x,\, \frac{1}{2}-y\right]$	[68]
69	$\left[y+\frac{1}{2},\, z,\, \frac{1}{2}-x\right]$	[69]
70	$\left[\frac{1}{2}-z,\, x,\, y+\frac{1}{2}\right]$	[70]
71	$\left[\frac{1}{2}-y,\, z,\, x+\frac{1}{2}\right]$	[71]
72	$\left[z+\frac{1}{2},\, -x,\, y+\frac{1}{2}\right]$	[72]
73	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[73]
74	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[74]
75	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[75]
76	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[76]
77	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y\right]$	[77]
78	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x\right]$	[78]
79	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, -x\right]$	[79]
80	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y\right]$	[80]
81	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x\right]$	[81]
82	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	[82]
83	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	[83]
84	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, -y\right]$	[84]
85	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[85]
86	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[86]
87	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[87]
88	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[88]
89	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, -y\right]$	[89]
90	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, -x\right]$	[90]
91	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, x\right]$	[91]
92	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, -y\right]$	[92]
93	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, -x\right]$	[93]

continued ...

Table 9

No.	position	mapping
94	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y\right]$	[94]
95	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x\right]$	[95]
96	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, y\right]$	[96]